Christoph B. Kassir

christoph.kassir@mail.utoronto.ca | [–Portfolio Site–] | linkedin.com/in/christophkassir

Education

University of Toronto 2023 – 2028
B.A.Sc. Electrical and Computer Engineering *Toronto, ON*

- UofT IEEE × Cognichip IC Hack **1st Place**, ECE295 × NWS Design Competition **1st Place**
- NSERC Research Award, First-Year Research Award, Dept & Faculty Entrance Scholarships, Dean's List
- Courses: Analog Data Converter ICs (**16nm FinFET**), VLSI Layout (**65nm CMOS**), Digital ICs (250nm CMOS), Fields & Waves, Power Electronics, Semiconductor Devices, Analog ICs, Digital Design, Computer Organization

Experience

Analog & Digital Layout Design Engineering Intern Incoming 2026
Marvell Technology *Toronto, ON*

- Designing high-speed layout (Virtuoso) in **2nm finFET & 3nm GAAFET** for Photonic Fabric™ RX SERDES.
- Built and presented an agentic LLM for automated generation of SKILL code for powergrid PCELLs and algorithms.
- Technology node pathfinding: device characterization and S-parameter simulations.

Hardware Engineering Intern May-Aug 2025
Myant Medtech *Toronto, ON*

- Full RTL design + verification for MHz-range FPGA converter controller (SystemVerilog, Quartus).
- Planar/2D transformer PCB design (Altium) and magnetic simulation (PLECS).
- Cramp-reducer PCB design (Altium): USB-C PD, DC-DC, impedance matching, OV/OC protection.
- Neural-diagnoser PCB design (Altium): digital delay circuits, DC-DC, OV/OC protection.
- Overhauled 2016 legacy firmware, rewrote BLE codebase in Zephyr-RTOS (C, Linux).

ML Research Intern Jan-Apr 2025
Department of National Defense (DND) *Toronto, ON*

- Built Hidden-Markov-tuned transformers (PyTorch) on an explainable-AI VIEWS25 contract.

Hardware Engineering Intern May-Aug 2024
University Health Network (UHN) *Toronto, ON*

- Updated smart-ring bare-metal SPI driver with DMA and interrupts (C): 5× speed-up.
- Built a genetic algorithm tool to fully automate DSP filter tuning (MATLAB) for sensor readings.

Projects

Neurocore Mixed-Signal IC: Analog IC + ASIC

- **Winner**: UofT IEEE × Cognichip IC Hack, 1st place.
- Custom ADC architecture (Virtuoso): asynch datapath, SRAM cells + timing, Spectre verification. **65nm CMOS**.
- ASIC (SystemVerilog): RTL-to-GDSII, time-shared DWT, LSK interface, power/clock gating. **130nm CMOS**.

CMOS EKV-based Simulator (ongoing project)

- CMOS simulator in C++17, emphasis on **moderate inversion** simulation using the EKV equations.

FPGA FFT: RTL vs HLS Comparison

- 8-point FFT unit for HLS vs RTL comparison (Vivado, HLS, SystemVerilog) + testbenches.
- Containerized solution (Bash, Docker) for Vivado installation on Apple Silicon.

PCB Wireline Audio Communication

- **Winner**: ECE295 × NWS Design Competition, 1st Place
- Bare-metal (C) for 4kHz A-D-A conversion on-PCB (Altium). PyVisa test-automated.